

Authority Circuit Standards Program

A Formal Architecture for Deterministic, Verifiable Public Decision Systems

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Publication Notice

This publication is a technical working draft. It is not legal advice, does not determine the validity of any real-world authority, and does not claim that every legal or public decision can be reduced to Boolean logic. It defines an engineering framework for making specified decision processes more observable, testable, reproducible, and correctable.

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Chapter 1

ACA-000 — Program Charter

1.1 Purpose

The Authority Circuit Standards Program defines open, modular specifications for observable public and organizational decision systems. Its purpose is to reduce dependence on unexplained authority by requiring stated inputs, explicit decision states, reproducible evaluation, proof artifacts, and operational remedy.

1.2 Design commitments

The program is governed by five commitments:

1. consequential decisions remain attributable to accountable human and institutional authority;
2. uncertainty is preserved rather than silently converted into approval or denial;
3. every decision is replayable from versioned rules and referenced evidence;
4. every detected error has an attached correction route;
5. published claims are separated from allegations, assumptions, and unresolved conflicts.

1.3 Publication model

Standards are developed openly through version control, issues, review, test fixtures, and immutable releases. Draft status SHALL be visible on every draft artifact.

Chapter 2

ACA-100 — Authority Circuit Architecture

2.1 System model

The architecture separates six layers: source authority, formal rule, evidence and provenance, deterministic evaluation, decision proof, and remedy execution.

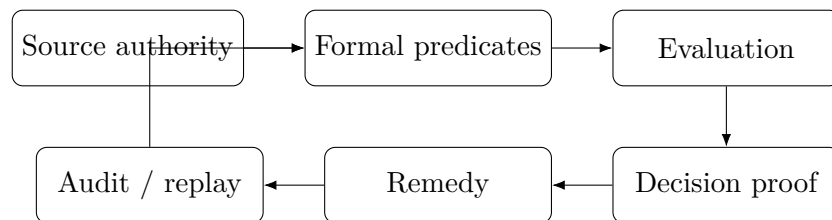


Figure 2.1: Closed-loop authority architecture.

2.2 Architectural invariants

ACA-100-001

A conforming implementation SHALL identify the versioned rule set used for every consequential decision.

ACA-100-002

A conforming implementation SHALL preserve unresolved, conflicting, expired, and revoked inputs as distinct machine-readable states.

ACA-100-003

A conforming implementation SHALL generate a Decision Proof Record for each completed evaluation.

ACA-100-004

A conforming implementation SHALL expose at least one defined remedy or review route for every non-final or adverse state.

Chapter 3

ACA-110 — Core Terminology

Authority object	Versioned representation of the claimed source, scope, delegate, jurisdiction, effective period, and revocation status of authority.
Evidence object	Referenced assertion or artifact with provenance, integrity metadata, access controls, and status.
Predicate	Typed proposition evaluated under a declared rule version.
Verifiable Authority Cell	Reusable decision primitive that evaluates mandatory predicates and emits a defined decision state.
Decision Proof Record	Portable record sufficient to inspect and independently replay an evaluation, subject to lawful access controls.
Operationalized remedy	Defined, time-bounded process for notice, review, correction, propagation, verification, and closure.
Authority Transistor	Public metaphor for the smallest reusable authority-control element; not a claim of a novel semiconductor device.

Chapter 4

ACA-120 — Verifiable Authority Cell

4.1 Canonical inputs

A baseline cell evaluates:

$$I = \{A, J, R, E, P, T, N, C, V\}$$

where A is authority validity, J jurisdiction, R rule applicability, E evidence sufficiency, P required procedure, T temporal validity, N notice and review, C conflict state, and V revocation or prohibition.

4.2 Allow expression

For profiles that permit Boolean authorization:

$$\mathbf{ALLOW} = A \wedge J \wedge R \wedge E \wedge P \wedge T \wedge N \wedge \neg C \wedge \neg V.$$

4.3 Required behavior

ACA-120-001

A mandatory predicate with value **UNKNOWN** SHALL prevent **ALLOW**.

ACA-120-002

A mandatory predicate with value **CONFLICTED** SHALL produce **CONFLICT** unless a profile defines a higher-priority safe state.

ACA-120-003

A valid revocation or prohibition SHALL take precedence over ordinary authorization.

ACA-120-004

Profiles SHALL identify which predicates are objective, evidentiary, discretionary, or non-computable. Human judgment SHALL NOT be concealed as deterministic computation.

Chapter 5

ACA-130 — Decision State Machine

5.1 Canonical states

$S \in \{\text{ALLOW}, \text{DENY}, \text{HOLD}, \text{CONFLICT}, \text{REVOKED}, \text{ESCALATE}\}.$

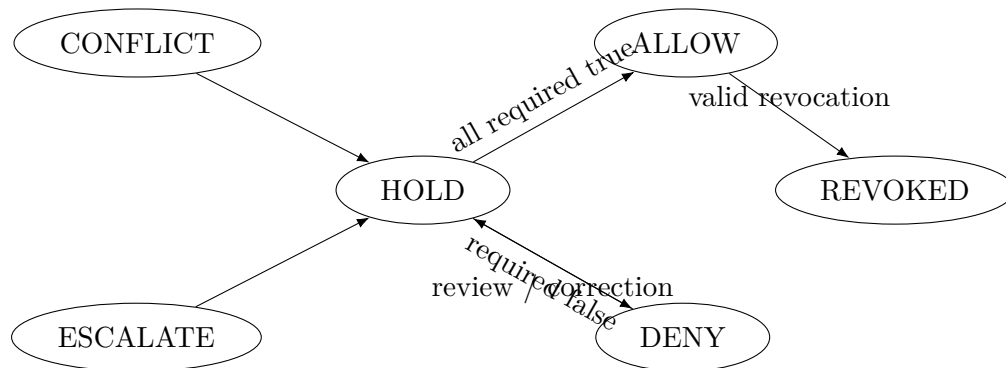


Figure 5.1: Baseline state transitions. Profiles may restrict transitions but may not erase provenance.

ACA-130-001

Every transition SHALL record its triggering event, actor or service identity, timestamp, prior state, resulting state, and rule version.

Chapter 6

ACA-140 — Decision Proof Record

6.1 Minimum record

A Decision Proof Record SHALL contain:

1. a unique record identifier;
2. the decision profile and rule version;
3. evaluation time and implementation identity;
4. each material predicate, value, source, and evidence reference;
5. the final state and transition reason;
6. identified conflicts, unknowns, expirations, and revocations;
7. the available remedy route and applicable deadline;
8. an integrity mechanism suitable to the deployment context.

ACA-140-001

The record SHALL support deterministic replay when the evaluator has lawful access to the same rule versions and evidence objects.

ACA-140-002

The record SHALL distinguish integrity proof from truth proof. A cryptographic hash proves that content has not changed; it does not prove that the original assertion was true.

Chapter 7

ACA-150 — Operationalized Remedy Protocol

7.1 Lifecycle

The baseline remedy lifecycle is:

Detect → Notify → Preserve → Review → Correct → Propagate → Verify → Close.

ACA-150-001

A remedy SHALL NOT be marked complete merely because a review occurred.

ACA-150-002

Closure SHALL require evidence that the corrective action was executed and, where applicable, propagated to dependent records and downstream decisions.

ACA-150-003

The protocol SHALL retain the original state, correction history, responsible authority, elapsed time, and verification result.

7.2 Core measures

Deployments SHOULD report: decision reproducibility rate, unresolved-input rate, contradiction-detection rate, median correction time, remedy completion rate, downstream propagation completeness, and independent replay success.

Chapter 8

Implementation Roadmap

8.1 First bounded profile

The first reference implementation should address a narrow decision with published rules, objective evidence, limited discretion, and a clear correction path. A filing-completeness or deadline-validity profile is preferable to a high-discretion adjudication.

8.2 Acceptance tests

The first implementation is not accepted until:

1. repeated evaluation of identical inputs produces identical records;
2. two independent implementations produce equivalent states and reasons;
3. missing evidence cannot produce **ALLOW**;
4. conflicting evidence is visibly preserved;
5. revocation propagates to dependent decisions;
6. every adverse or incomplete result identifies a testable remedy route.

8.3 Strategic outputs

The program should publish, in order: architecture, terminology, cell specification, state machine, proof schema, remedy protocol, test fixtures, reference implementation, independent evaluation, and conformance profile.

Appendix A

Minimal Decision Proof Record

```
{
  "record_id": "DPR-example-0001",
  "state": "HOLD",
  "predicates": [
    {"id": "authority.valid", "value": "TRUE"},
    {"id": "evidence.complete", "value": "UNKNOWN"}
  ],
  "remedy": {"status": "AVAILABLE", "route": "supply-missing-evidence"}
}
```